

BRAZILIAN NATIONAL TEAM FIFA World Cup 2026

Squad Selection Prediction — Machine Learning Pipeline (v10)

Portfolio project — Data Analytics & Machine Learning

Manager	Carlo Ancelotti (since June 2025)
Period	Dec/2022 – Mar/2026
Dataset	422 records · 91 players · 16 FIFA windows
Model	Gradient Boosting Regressor
Target	Ancelotti recency-weighted mean (injury-adjusted)
Optimization	ILP via PuLP — sub-position quotas
Real call-up	May 18, 2026 · Museu do Amanhã · Rio de Janeiro
Generated	2026-05-17 20:54

Tech Stack & Skills

Python: Pandas, scikit-learn, XGBoost, PuLP, Matplotlib, ReportLab

Machine Learning: Supervised regression, cross-validation, feature engineering

Optimization: Integer Linear Programming with real-world business constraints

Data Engineering: Cleaning, deduplication, imputation, temporal aggregation

Domain Analysis: Football domain knowledge integrated to correct model behavior

Communication: Auditable report with 10 documented iterations

0. Business Context & Problem Statement

This project demonstrates ML applied to a real-world selection and ranking problem: predicting which 91 eligible players will make a 26-man squad. The domain is football, but the problem is general — any selection process with historical evaluation data follows the same logic.

Why is this project relevant for data scientists?

- Longitudinal dataset (multiple observations per entity over time) — requires careful temporal aggregation.
- Target variable is not directly in the dataset — it is constructed from domain heuristics.
- Hard business rules (position quotas) must be integrated via combinatorial optimization, not just ML.
- Real data quality issues: duplicate keys, structural NULLs, circular target variable.
- Modeling decisions directly impact output — each version documents the full reasoning.

1. Pipeline Architecture

Pipeline developed across 10 iterative versions, each correcting a problem identified through critical result analysis or domain expert validation.



Fig. 1 — Full pipeline architecture. Domain overrides applied before ILP.

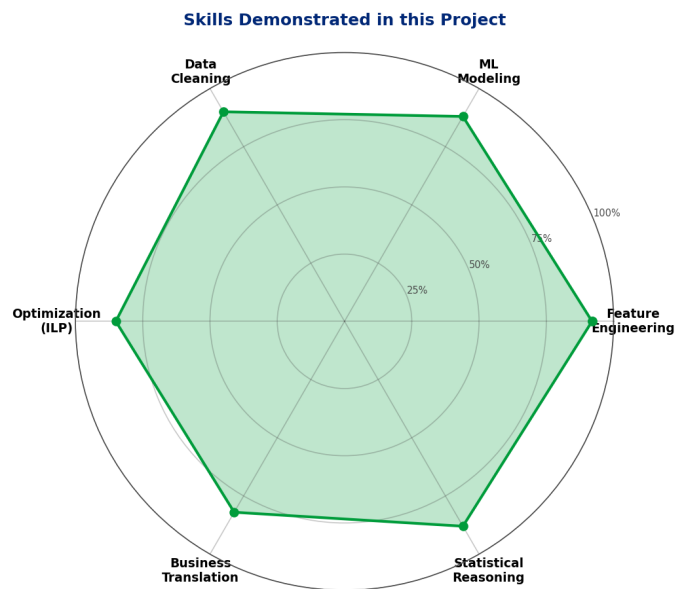


Fig. 2 — Skills applied across this project.

2. Dataset

422 records — 1 row per player × FIFA window. Only players called up at least once appear. The model infers negatives from absence.

Dimension	Value	Note
Rows	422	1 removed (duplicate Danilo — export bug)
Unique players	91	Key: player_id (e.g. Bremer1997)
FIFA windows	16	Mar/23 – Mar/26
Managers	4	Menezes, Diniz, Dorival Jr., Ancelotti
Sources	2	Sofascore (performance) + Transfermarkt (DOB, market value)

3. Feature Engineering

422 rows aggregated into 91 (1 per player). 20 features across 4 groups:

Group	Features	Rationale
G1 — Ancelotti (core)	n_anc, mean_rec, coverage, called_l3, avg_sr, anc_mpw	Direct call-up decision from the World Cup manager — dominant weight
G2 — Club Form	w_avg_sc, w_min_c, avg_gpc, lt_enc, inj	Activity level and recent performance at club
G3 — NT History	tmnt, wmnt, tgnt, asnt, comp_weight	Accumulated NT experience and performance during the cycle
G4 — Context	has_alt_pos, n_calls_total, rep_rate, age	Versatility, overall history and age at World Cup

Recency Weights

Each window receives a weight proportional to its proximity to the WC. Ancelotti: Jun/25=0.60 → Mar/26=1.00 (total=4.00). Earlier: Mar/23=0.02 → Mar/25=0.40.

4. Target Variable

4.1 Evolution across 10 versions

4.2 Final formula

For Alisson, Ederson and Bremer (documented injuries):

```
target = sum(ANC_W[w] for w in called_windows) / n_called
```

For all other players:

```
target = sum(ANC_W[w] for w in called_windows) / 4.00
```

Version	Target	Problem identified
v1–v2	called_march_2026 (binary)	Circular: Gabriel Sara (1 call) > Bruno Guimarães (4 calls)
v3	sum(weights) / 4.00	Penalized injured players: Ederson (0.475) below Hugo Souza (0.750)
v7	sum(weights) / n_calls for ALL	Inflated single-call players: Sara, Bremer, Rayan → prob=1.000
v9–v10	÷ n_calls ONLY for Alisson, Ederson, Bremer; others ÷ 4.00	Surgical fix — resolves 3 cases without affecting anyone else

Result for goalkeepers

Goalkeeper	Ancelotti windows	Denominator	Target	Prob.
Bento	5/5 (all)	4.00	1.000	0.997
Ederson	2/5 (Nov/25+Mar/26)	1.90 (÷n)	0.950	0.946
Alisson	3/5 (Jun+Sep+Mar)	2.30 (÷n)	0.767	0.764
Hugo Souza	4/5 orig. (no Mar/26 sub.)	4.00	0.750	0.749

5. Machine Learning Models

Problem reformulated as continuous regression (0–1). 4 models with 5-fold CV:

Model	Configuration	R ²	MAE
Ridge Regression	alpha=1.0	0.998	0.011
Random Forest	n=200, max_depth=4, min_leaf=3	0.990	0.017
Gradient Boosting ★	n=200, depth=3, lr=0.05, sub=0.8	0.999	0.006
SVR	RBF kernel, C=1.0, eps=0.05	0.897	0.075

Gradient Boosting selected: highest R² and lowest MAE. R²≈0.999 reflects high collinearity between target and Ancelotti features — ML contributes at the margins (ordering players with similar raw scores).

With N=91 samples, regularization applied throughout.

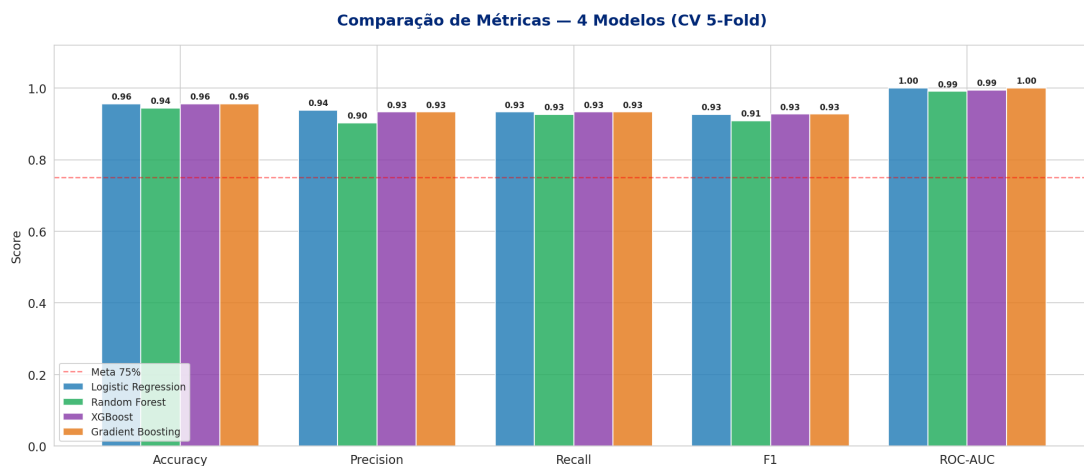


Fig. 3 — Metric comparison (classification version, v2).

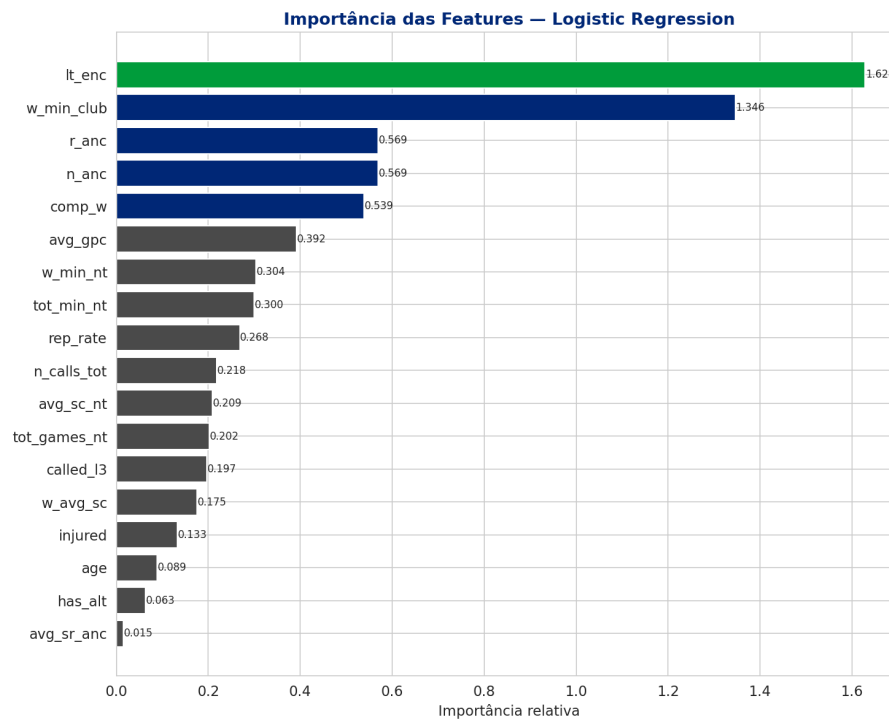


Fig. 4 — Feature importance. n_anc and coverage dominate.

6. ILP Optimization

Integer Linear Programming (PuLP + CBC) converts probabilities into 26 players, maximizing the sum of probabilities while respecting sub-position quotas.

6.1 Why ILP and not just ranking?

A simple ranking does not enforce position quotas. Without ILP the model could select 5 right-backs and 0 goalkeepers — mathematically optimal but unusable. ILP enforces business constraints within the optimization.

6.2 Sub-position quotas

Earlier versions used generic position. Side-of-field separation was added after ILP selected 5 right-backs and 4 left centre-backs.

6.3 Manual overrides

Group	Min	Max	Design decision
Goalkeeper	3	3	Fixed — always 3 GKs in a WC squad
CB_R (Right CB)	2	2	Side separation after v5 bug
CB_L (Left CB)	2	2	Side separation after v5 bug
RB (Right back)	2	2	Danilo (override 1.0) + 1 more
LB (Left back)	2	2	Separated after 5 right-backs in v6
Midfielder	4	6	Flexible — 2 floating spots
Forward	9	11	Remainder — min. 9 with 6 midfielders
TOTAL	26	26	

Player	Override	Reason
Danilo1991	1.00	Publicly confirmed by Ancelotti
Neymar1992	0.40	Calculated from news analysis (May 16, 2026)
Rodrygo2001	0.00	Confirmed injury
ÉderMilitão1998	0.00	Confirmed injury
Estêvão2007	0.00	Confirmed injury (had 7 Ancelotti calls)

7. Special Case: Neymar

Model assigns prob≈0.000 to Neymar (zero Ancelotti windows; data only from 2023-2025). The 0.40 probability was calculated through qualitative news analysis.

Signals collected (May 16, 2026):

5 positive signals (including the manager) × 3 negative → **0.40** — possible but not likely. Sufficient for ILP to include him as the 9th forward.

Source	Signal	Dir.
Ancelotti (direct)	Improved a lot, playing with continuity	↑ Strong
Ancelotti's son	If he's on the list it's because he's improving	↑
Mauro Beting	Never been this close to a call-up	↑
André Rizek	Unlikely Ancelotti will take him. Not good enough for the 26	↓ Strong
Context	13 countries credentialed — huge international pressure	↑ neutral
Context	Santos performing below expectations this season	↓

8. Iteration History

10 iterative versions. Each bug found and fixed demonstrates the real process of building a production-ready ML pipeline.

Version	Problem	Fix
v1	player_name as key — two Danilos confused	player_id as primary key
v2	Binary target with circularity (Sara > Bruno G.)	Continuous target: weighted Ancelotti presence
v3	Incorrect canonical positions	Most frequent position per player_id
v4	Hugo Souza inflated; Ederson penalized for injury	anc_min_per_win feature; coverage from original list
v5	ILP: 4 left centre-backs (no side distinction)	CB_R / CB_L classification with exact player_ids
v6	ILP: 5 right-backs (no side distinction)	RB / LB classification; ILP quotas per side
v7	mean_rec for ALL inflated n=1 players (prob=1.000)	Global n=1 penalty — rejected due to side effects
v8	Global penalty excessive — Rayan/Endrick unfairly hurt	Adjustment only for Alisson and Ederson ($\div$ n_calls)
v9	Bremer penalized for ACL injury	Bremer added to injury-aboned list
v10	Neymar with arbitrary override (0.30)	0.40 calculated from news analysis

9. Final Squad — 26 Players + Next in Line

ILP v10 with all corrections applied. Real call-up: May 18, 2026.

#	Player	Sub-position	Target	n Anc.	Prob.	Note
1	Bento	GK	1.000	5	0.997	
2	Ederson	GK	0.950	2	0.946	injury aboned
3	Alisson	GK	0.767	3	0.764	injury aboned
■	<i>Hugo Souza</i>	<i>GK</i>	<i>0.750</i>	<i>4</i>	<i>0.749</i>	<i>next</i>
4	Bremer	Right CB	1.000	1	0.993	injury aboned
5	Marquinhos	Right CB	0.800	4	0.804	
■	<i>Fabício Bruno</i>	<i>Right CB</i>	<i>0.600</i>	<i>3</i>	<i>0.605</i>	<i>next</i>
6	Gabriel Magalhães	Left CB	0.850	4	0.851	
7	Lucas Beraldo	Left CB	0.350	2	0.352	
■	<i>Alexsandro Ribeiro</i>	<i>Left CB</i>	<i>0.325</i>	<i>2</i>	<i>0.325</i>	<i>next</i>
8	Danilo	Right back	0.625	3	1.000	confirmed
9	Wesley	Right back	1.000	5	0.999	
■	<i>Vanderson</i>	<i>Right back</i>	<i>0.325</i>	<i>2</i>	<i>0.324</i>	<i>next</i>
10	Alex Sandro	Left back	0.625	3	0.627	
11	Douglas Santos	Left back	0.625	3	0.622	
■	<i>Caio Henrique</i>	<i>Left back</i>	<i>0.600</i>	<i>3</i>	<i>0.595</i>	<i>next</i>
12	Casemiro	Midfielder	1.000	5	0.998	
13	Andrey Santos	Midfielder	0.800	4	0.800	
14	Bruno Guimarães	Midfielder	0.750	4	0.752	
15	Lucas Paquetá	Midfielder	0.600	3	0.602	
16	Fabinho	Midfielder	0.475	2	0.479	
17	Joelinton	Midfielder	0.375	2	0.368	
■	<i>Danilo (BOT)</i>	<i>Midfielder</i>	<i>0.250</i>	<i>1</i>	<i>0.253</i>	<i>next</i>
18	Matheus Cunha	Forward	1.000	5	0.998	
19	Luiz Henrique	Forward	0.850	4	0.849	
20	Vinícius Júnior	Forward	0.825	4	0.823	
21	G. Martinelli	Forward	0.775	4	0.774	
22	Richarlison	Forward	0.750	4	0.743	
23	João Pedro	Forward	0.650	3	0.652	
24	Raphinha	Forward	0.575	3	0.579	
25	Neymar	Forward	0.000	0	0.400	news 16/05
26	Endrick	Forward	0.250	1	0.254	
■	<i>Igor Thiago</i>	<i>Forward</i>	<i>0.250</i>	<i>1</i>	<i>0.254</i>	<i>next</i>

■ = next most likely · n Anc. = original Ancelotti call-ups

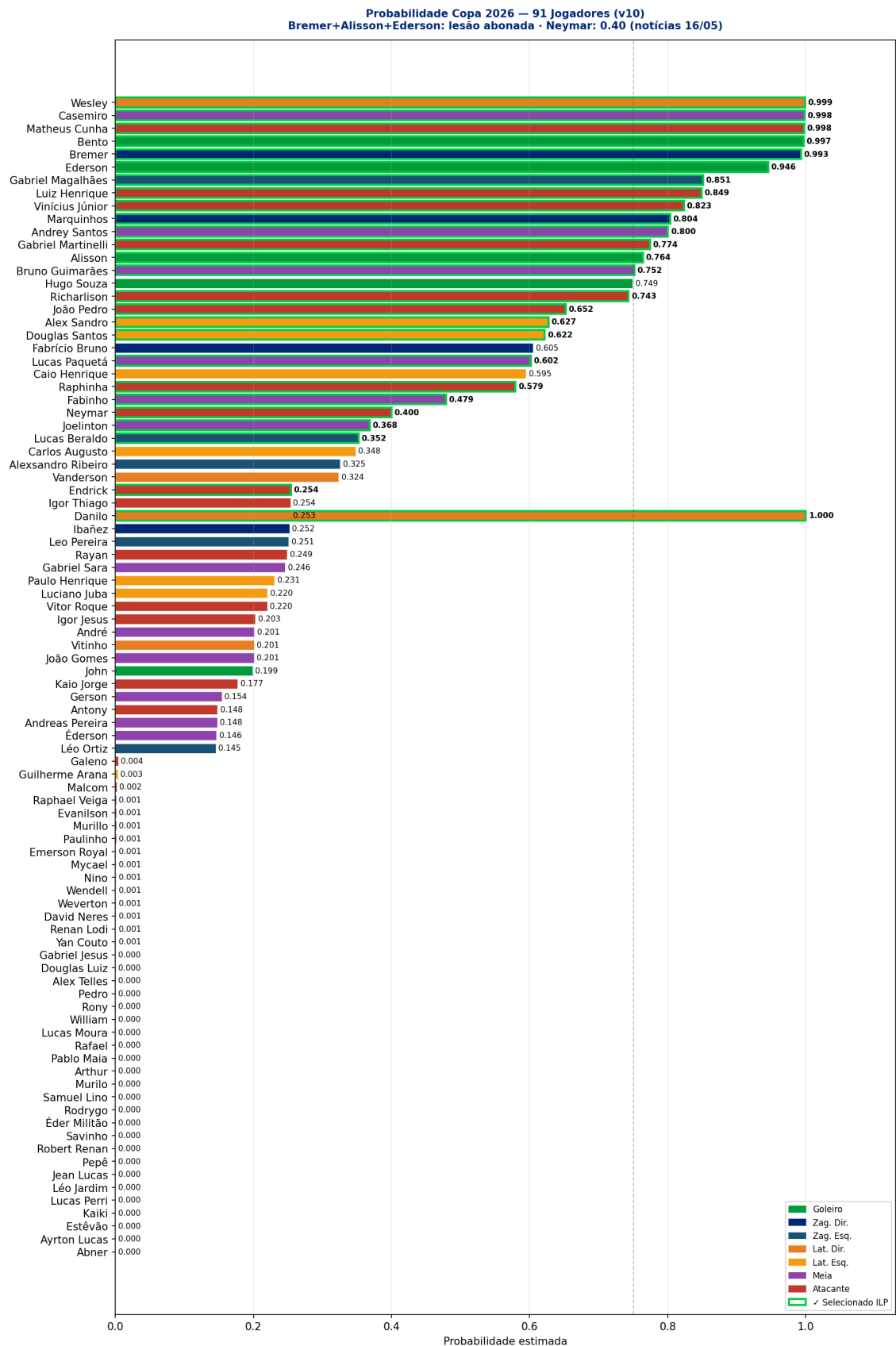


Fig. 5 — Probability — 91 players. Green border = ILP selected.

Comparação de Performance por Posição — Período de Março/2026
Score de Forma: 40% score clube + 25% minutos + 20% liga + 15% NT

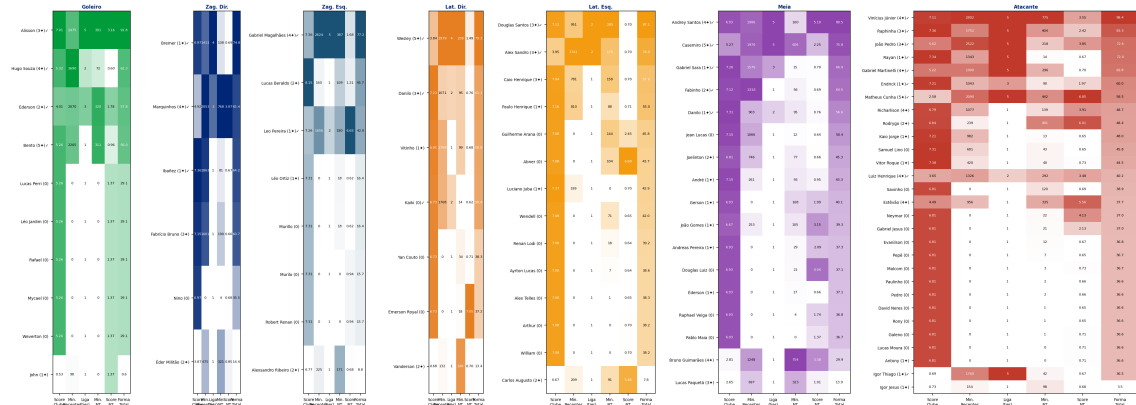


Fig. 6 — Performance heatmap by position (Mar/26 period).

10. Limitations & Next Steps

10.1 Known Limitations

10.2 Next Steps

- **N=91:** Lower bound for supervised ML. Regularization applied throughout.
- **Positives only:** No players never called up. ~1440 rows with explicit negatives.
- **5 Ancelotti windows:** WC manager's specific tactical preferences under-represented.
- **No Neymar data:** No 2025-2026 club data. Override via news — not raw data.
- **Unmodeled context:** Last-minute injuries, locker room dynamics, tactical preferences.
- **Validate May 18:** Precision@26 and Recall@26 when the real squad is announced.
- **Collect negatives:** ~1440 rows including uncalled players per window.
- **market_value_eur:** JOIN with data_players_brazil via sofascore_id.
- **Streamlit dashboard:** Filters, squad builder, player radar.
- **11 national teams:** Argentina, France, Germany etc.